

PROFESSIONAL SUMMARY

Motivated Electrical and Electronic Engineering graduate, currently pursuing an MSc in Electrical Engineering and Information Technology at ETH Zürich, with strong academic results and practical experience in digital design, verification, and embedded systems. Completed a year-long placement at Arm, supporting pre-silicon verification, timing analysis, and RTL debugging.

Interested in digital hardware development across the design, verification, and implementation flow, as well as FPGA development, embedded systems, and low-level programming.

EMPLOYMENT HISTORY

- ❖ CPU Physical Implementation Intern, Arm Jun 2024 — Jun 2025
Cambridge, UK

Supported digital design and verification flows for CPU IP, including development of automation scripts, timing and power analysis, and debugging RTL/netlist mismatches. Contributed to test-plan development, regression simulations, and coverage analysis to ensure robust pre-silicon checks. Collaborated across verification and physical design teams to streamline methodology improvements.
- ❖ Research Intern , Allen & Heath Jul 2026 — Aug 2026
Penryn, UK

Developed and verified a real-time SMPTE Linear Time Code (LTC) decoder, progressing from Python algorithm development to synthesizable VHDL RTL. Implemented and validated the design on a Digilent Zybo FPGA-SoC, including a custom I2S audio receiver. Developed a software-based decoder for comparison, using DMA-based audio transfer between Programmable Logic and the Cortex-A9 Processing System. Verified end-to-end decoding by streaming recovered LTC data to the Processing System for UART display.
- ❖ PALs leader, University of Plymouth Sept 2023 — Apr 2024
Plymouth, UK

Organized and led Peer Assisted Learning sessions for first-year students, providing academic support and guidance.
- ❖ Project Support and Development Micro-Internship, ConeX Portal Jul 2023 — Aug 2023
UK

Researched industrial clusters and professional platforms; proposed strategies for new product features while gaining professional workplace experience.
- ❖ Math and Physics tutor, self employed May 2022 — May 2024
Plymouth, UK

Provided personalized support for school-level students, reinforcing subject mastery while developing communication skills.

EDUCATION

- ❖ University of Plymouth Sept 2022 — Jun 2026
Plymouth

BEng Electrical and Electronic engineering

I was awarded the Dean's Prize for Best Overall Performance by an International Student for my overall grades of 91.39%.

I was also awarded the Tony Rees Memorial Award for the Best Application of Technological Skills for my final year project

I am also the IET Prize winner.

Modules taken and semester-end results,

Stage 3:

- Final Year Individual Project - 91.58%
- ELEC345 (High Speed Communications) - 95.6%
- ELEC347 (Information and Communication Signal Processing) - 98%
- ELEC351 (Advanced Embedded Programming) - 77.6%
- ELEC349 (Design and Control of Renewable Energy Technology) - 92%

Stage 2 :

- ROCO219 (Control Engineering) - 92.6%
- PROJ200 (Real Time Systems Project) - 96%
- MATH237 (Engineering Mathematics and Statistics) - 92.85%
- ELEC239 (Communication Systems) - 85.7%
- ELEC237 (Power Electronics and Generation) - 89.8%

Stage 1 :

- ELEC144 (Electrical and electronic principals) - 83.4%
- ELEC143 (Embedded systems) - 84%
- MATH190 (Engineering mathematics) - 88.74%
- ELEC141 (Analogue Electronics) - 98%
- ELEC142 (Digital Electronics) - 92.4%
- PROJ101 (Electronic and Robot Design Build team project) - 78.8%

❖ **Musaeus College** Oct 2020 — May 2022
A level Colombo

4 A*'s in science field subjects, Physics, Maths, Chemistry and Biology, and received the outstanding Cambridge learner award.

❖ **Visakha Vidyalaya** Jan 2012 — Oct 2020
O level Colombo

5 A*'s in English, Maths, Physics, Chemistry and Biology.

Bronze medal at the Sri Lankan Junior Science Olympiad Competition - 2019

SKILLS

Embedded programming: baremetal
through to RTOS

Programming & Scripting: C, C++,
Python, Bash (Linux), TCL.

HDL & Design: SystemVerilog,
VHDL.

Verification & Simulation:
SystemVerilog, RTL simulation, debug
(waveform analysis).

Tools & Methodologies: Cadence,
Synopsys, MATLAB/Simulink, FPGA
prototyping.

Other: Hardware functional
simulation, benchmarking, test-plan
development.

LANGUAGES

English *Highly proficient* Sinhala *Native speaker*

PROJECTS

❖ **Embedded DSP Acceleration: A space-time journey with flip-flops and butterflies** Dec 2025 — May 2026

- This was my final year project
- This project takes a block floating point 8-point Fast Fourier Transform (FFT) core from concept to reality-optimizing its power, performance, and area on an FPGA, then tying it into a Raspberry Pi with high-speed SPI.
- It uses a custom driver to abstract protocol, and benchmarks the hardware against a straightforward software implementation.
- It explores digital design concepts such as pipelining, hardware reuse and clock domain crossing

❖ **Biquad Yagi Antenna & PCB with 433 MHz Antenna and RF gain stage** Sept 2025 — Dec 2026

- Investigated the design, fabrication, and characterisation of a self-selected and designed Biquad-Yagi antenna
- A provided 433 MHz antenna and an RF gain stage were characterized
- The impedance matching circuit was designed and the PCB fabricated
- With the self-designed antenna, the primary objective was to achieve improved directivity using a Biquad-driven Yagi structure and experimentally validate RF behaviours such as S parameters, impedance matching, link budget performance, and 3D radiation patterns.
- This data was used to complete the project by creating a datasheet for the Biquad-Yagi antenna

❖ **Heart Rate Monitor** Apr 2024

- Designed and implemented an end-to-end heart rate monitoring system, including analogue front-end and FPGA sampling.
- Developed HDL testbenches for SPI and LCD modules, validating functionality through simulation before hardware deployment.
- Applied systematic debugging and functional verification, comparing results against commercial fitness trackers to ensure design correctness.

❖ **State Feedback Control of an Inverted Pendulum** Apr 2024

- Implemented digital control algorithms for stabilising an inverted pendulum, achieving both physical and simulated operation.
- Demonstrated ability to move from simulation validation to hardware testing, ensuring consistent performance.

❖ **Ultrasound Communications System** Oct 2023

- Designed and tested a wireless ultrasound link for digital communication.
- Verified modulation/demodulation logic by running simulations under varied channel conditions and baud rates.
- Performed systematic error analysis and coverage of test scenarios, identifying edge cases and validating design robustness.

❖ **Bead-pushing robot** Mar 2023

- Group project to design a robot capable of transporting beads across an arena.
- Developed micro-switch detection for ramp and wall navigation.
- Learned the importance of early integration and testing after encountering last-minute hardware/software compatibility issues.

❖ Digital Level Dec 2022

- Built using Nucleo F409RE and MPU6050 sensor, with real-time data displayed on an LCD and LED matrix.
- Learned the value of thorough testing after losing marks for insufficient corner-case validation.

❖ Combat bots Dec 2022

- Designed and built a servo-powered lifter bot controlled via radio.
- Introduced to CAD (Fusion 360), designing chassis and components within strict dimensional constraints.

❖ T-rex (chrome no internet) game Oct 2022

- Implemented using Nucleo F409RE, LCD, and custom button input.
- Applied embedded systems knowledge to create a playable game with custom characters.

H O B B I E S

❖ Art

Hyper-realistic drawing, Watercolor

❖ Music

Guitar, Singing, Violin

❖ Sports

Running, Hiking, Badminton, Ultimate Frisbee